

LOUISA COUNTY AP, VA, USA

WMO: 723114

Lat: 38.010N

Lon: 77.970W

Elev: 493

StdP: 14.44

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 03715

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	13.8	18.7	-2.2	5.0	23.6	2.6	6.4	25.1	20.3	41.7	18.2	42.2	3.2	280	0.386

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.5	94.5	77.6	91.3	76.7	90.1	76.2	81.4	87.8	79.9	87.1	78.3	86.3	4.7	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.5	155.7	84.6	78.6	151.3	84.2	75.4	135.6	81.7	45.5	87.6	43.7	87.3	42.3	87.4	85.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
16.8	13.5	11.4	DB	5.2	97.2	5.5	3.2	1.2	99.5	-2.0	101.4	-5.1	103.3	-9.1	105.6	
			WB	4.6	82.9	5.4	2.1	0.8	84.4	-2.4	85.7	-5.4	86.9	-9.3	88.4	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	58.2	37.2	40.6	48.2	58.6	66.5	73.9	77.8	76.6	70.0	59.0	48.5	40.9
	DBStd	16.51	11.09	10.18	10.55	9.16	7.53	5.78	4.91	5.04	6.26	8.57	9.35	9.37
	HDD50	1378	422	295	167	23	0	0	0	0	0	20	140	311
	HDD65	3889	863	683	527	235	76	5	0	0	26	226	501	747
	CDD50	4387	25	32	111	282	512	717	862	823	601	299	93	30
	CDD65	1424	1	1	6	44	123	272	397	358	177	40	4	1
	CDH74	13727	3	11	91	547	1189	2654	4027	3386	1443	336	37	3
	CDH80	5518	0	1	11	182	375	1075	1847	1451	487	83	6	0
(14) Wind	WSAvg	4.2	4.7	5.1	5.5	5.4	4.1	3.5	3.1	2.9	3.4	3.8	4.2	4.3
(15) Precipitation	PrecAvg	45.0	3.1	2.8	4.1	3.1	4.2	4.0	4.4	3.7	4.4	3.8	3.4	3.3
	PrecMax	67.1	6.5	8.2	8.9	6.4	8.2	7.4	8.7	6.9	10.2	9.1	7.4	7.2
	PrecMin	36.8	1.1	0.7	0.2	1.1	1.1	1.0	1.3	1.1	0.6	0.2	0.3	1.5
	PrecStd	7.3	1.4	1.7	2.0	1.2	2.0	1.8	2.0	1.4	2.8	2.5	2.0	1.6
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	71.7	75.0	81.4	90.2	90.4	95.5	99.2	97.3	92.8	87.8	79.3	72.1
		MCWB	61.8	61.6	62.5	65.8	74.1	78.1	77.3	79.0	74.7	73.8	65.0	61.8
	2%	DB	64.9	68.2	75.4	85.6	87.9	91.4	94.9	93.2	89.7	81.8	72.6	65.5
		MCWB	58.0	56.9	59.3	65.1	73.0	77.0	77.8	77.5	75.3	69.1	62.5	60.5
	5%	DB	61.0	63.2	71.8	81.2	84.0	90.0	91.4	90.8	85.5	78.0	68.3	61.3
		MCWB	53.0	52.1	57.4	63.9	71.4	76.5	77.2	76.9	73.2	67.1	60.4	55.2
	10%	DB	54.5	57.2	66.1	75.2	81.4	86.5	90.0	88.2	81.9	73.1	64.0	56.5
		MCWB	47.6	48.4	55.7	61.6	69.8	74.8	77.0	75.9	71.4	65.9	57.0	49.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	64.0	63.8	66.4	72.0	78.1	82.4	83.0	83.5	80.8	75.7	70.5	66.7
		MCDB	67.5	71.1	74.6	82.2	86.2	89.0	89.9	91.0	85.1	82.3	73.1	69.1
	2%	WB	59.9	59.6	63.3	68.5	75.3	80.0	81.2	81.6	77.3	72.8	65.4	61.7
		MCDB	64.4	65.6	72.3	78.4	83.0	87.1	87.7	88.6	85.0	79.0	70.0	64.0
	5%	WB	54.0	54.4	60.3	66.3	73.4	78.1	79.7	79.8	75.3	70.4	62.0	56.4
		MCDB	58.2	60.4	68.4	75.8	82.1	86.7	88.0	86.2	81.5	74.9	66.8	60.2
	10%	WB	48.4	49.6	57.1	63.9	71.5	76.3	78.2	77.8	73.7	67.3	58.6	51.3
		MCDB	54.2	57.0	65.0	72.1	79.2	84.2	87.0	85.0	79.2	72.4	63.6	54.3
(35) Mean Daily Temperature Range	5% DB	MDBR	19.1	20.9	22.7	23.6	22.0	21.1	20.5	20.4	20.5	22.1	21.9	18.9
		MCDBR	24.1	27.0	28.2	29.0	25.1	24.1	23.4	23.3	25.2	25.7	25.8	22.9
	5% WB	MCWBR	17.7	18.0	16.0	12.9	12.8	11.7	9.8	9.8	12.1	14.2	15.9	17.2
		MCWBR	21.4	24.1	24.4	23.9	21.6	21.2	19.9	19.2	19.6	20.0	20.7	19.9
(40) Clear-Sky Solar Irradiance	taub		0.296	0.312	0.347	0.387	0.429	0.479	0.499	0.502	0.407	0.351	0.317	0.301
		taud	2.523	2.464	2.413	2.320	2.222	2.127	2.095	2.066	2.318	2.470	2.523	2.544
	Ebn at Noon		284	292	290	283	272	257	251	246	266	274	273	274
		Edh at Noon	24	30	34	40	45	49	50	51	37	29	24	23
(44) All-Sky Solar Radiation	RadAvg		712	962	1276	1593	1790	1962	1880	1692	1392	1082	801	625
	RadStd		59	109	88	122	165	135	102	82	157	124	62	56

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (72 neighbors)		+0.74	N/A	N/A	N/A	N/A	N/A	N/A	+241	+144

Nomenclature: See separate page